Supplementary Material: Uncertainty-Aware Multi-Agent Ransomware Detection via Decision MLP Fusion, Score-Space Augmentation, and Bayesian Triage Policy Optimisation

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This document provides supplementary material for the above paper. Section numbers use the prefix "S-" to distinguish them from the main manuscript. All equation, figure, and table numbers continue from the main paper.

I. Behavioural Feature Extraction Algorithm

Algorithm 1 describes the 16-dimensional feature extraction procedure from the raw RanSMAP CSV telemetry files. Each CSV row corresponds to a single I/O or memory event captured by the BitVisor hypervisor [1]. Features are computed over non-overlapping T_{window}=10 s windows.

Algorithm 1 16-dimensional behavioural feature extraction

Require: CSV event stream $E = \{e_k\}_{k=1}^{n}$; window size T_w=10 s
Ensure: Feature matrix $X \in \mathbb{R}^{W\times 16}$, one row per window

Partition E into W non-overlapping windows of duration T_w
for each window $w = 1, \ldots, W$ do
  $E_w \leftarrow \{e_k : t(e_k) \in [(w{-}1)T_w,\ wT_w)\}$
  // Storage features
  $\bar{T}_{s,r} \leftarrow \sum_{\text{read}} \text{bytes}/T_w$; $\bar{T}_{s,w} \leftarrow \sum_{\text{write}} \text{bytes}/T_w$
  $V_{s,r} \leftarrow \text{Var}(\{\text{LBA}(e) : e \in E_w,\ \text{type}(e){=}\text{read}\})$
  $V_{s,w} \leftarrow \text{Var}(\{\text{LBA}(e) : e \in E_w,\ \text{type}(e){=}\text{write}\})$
  $\bar{H}_{s,w} \leftarrow \text{mean}_{e\in E_w,\,\text{type}(e)=\text{write}} H(\text{sector}(e))$
  $\chi_s \leftarrow [\bar{T}_{s,r},\ \bar{T}_{s,w},\ V_{s,r},\ V_{s,w},\ \bar{H}_{s,w}]$ ▷ 5D storage
  // Memory features
  $H_{m,w} \leftarrow H(\{\text{GPA}(e) : e \in E_w,\ o(e){=}w\})$
  $H_{m,rw} \leftarrow H(\{\text{GPA}(e) : e \in E_w,\ o(e) \in \{r, w\}\})$
  For each $(s, o) \in \{4K, 2M, \text{MMIO}\} \times \{r, w, rw, x\}$: compute $C_{s,o}$ as event count
  $V_{m,w} \leftarrow \text{Var}(\{\text{GPA}(e) : e \in E_w,\ o(e){=}w\})$
  $\chi_m \leftarrow [H_{m,w},\ H_{m,rw},\ C_{4K,r},\ C_{4K,w},\ C_{4K,rw},\ C_{4K,x},\ C_{2M,r},\ C_{2M,rw},\ C_{\text{MMIO},r},\ C_{\text{MMIO},w},\ V_{m,w}]$ ▷ 11D
  $X[w, :] \leftarrow [\chi_s\,;\,\chi_m]$ ▷ 16D feature row
end for
return X

II. Decision MLP Training Procedure

Algorithm 2 describes the full Decision MLP training procedure including CTGAN augmentation, temperature calibration, and Bayesian TPE policy optimisation.

Algorithm 2 Decision MLP training with CTGAN augmentation and Bayesian TPE

Require: Trained agents $\{f_i\}_{i=1}^{3}$; labelled score-pool splits $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}$; N_{synth}=2,000, λ=0.3

// Stage 1: Score-pool construction
for each split $D \in \{\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{test}}\}$ do
  Compute $\mathbf{v}_j = (p_{\text{sem},j}, \sigma_{\text{sem},j}, p_{\text{beh},j}, \sigma_{\text{beh},j}, p_{\text{know},j}, \sigma_{\text{know},j})$ for each sample j
end for
// Stage 2: CTGAN augmentation (training split only)
Fit CTGAN_1 on $\{(\mathbf{v}_j, 1) : y_j{=}1,\ j \in D_{\text{tr}}\}$ for 50 epochs
Fit CTGAN_0 on $\{(\mathbf{v}_j, 0) : y_j{=}0,\ j \in D_{\text{tr}}\}$ for 50 epochs
$\mathcal{D}_{\text{syn}} \leftarrow \text{CTGAN}_1.\text{sample}(N_{\text{synth}}/2) \cup \text{CTGAN}_0.\text{sample}(N_{\text{synth}}/2)$
Clip $\mathcal{D}_{\text{syn}}$ to $[0, 1]^6$; remove NaN rows
$\mathcal{D}_{\text{aug}} \leftarrow \mathcal{D}_{\text{tr}} \cup \mathcal{D}_{\text{syn}}$
// Stage 3: Decision MLP training
Fit StandardScaler on $\mathcal{D}_{\text{aug}}$
Initialise MLP g_ϕ (6→32→1, Dropout p=0.2)
Train g_ϕ on $\mathcal{D}_{\text{aug}}$ with Adam $\eta{=}10^{-3}$, early stopping on $\mathcal{D}_{\text{val}}$ loss (patience = 6)
Restore best weights
// Stage 4: Temperature calibration
$T^* \leftarrow \arg\min_{T>0} \text{ECE}(g_\phi(\mathbf{v}/T);\ \mathcal{D}_{\text{val}})$ via L-BFGS
$\hat{p}_{\text{cal}} \leftarrow \sigma(g_\phi(\mathbf{v})/T^*)$
// Stage 5: Bayesian TPE policy optimisation
Run 300 Optuna TPE trials on $\mathcal{D}_{\text{cal}}$: $\Theta^* \leftarrow \arg\max_\Theta J(\Theta) = \text{bal_acc}(\Theta) - \lambda \cdot \text{esc_rate}(\Theta)$
return g_ϕ, T^*, Θ^*

III. Confusion Matrices and Per-Class Accuracy

Figure 1 shows normalised binary confusion matrices for the Behavioural CNN on the random-stratified, family-holdout, and hardware-holdout test splits. Figure 2 shows per-class accuracy for all twelve classes (six ransomware families, six benign applications) across the same three splits.

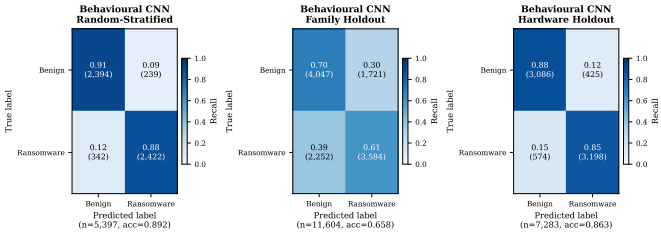


Fig. 1. Normalised binary confusion matrices for the Behavioural CNN (MC Dropout T=10) on three evaluation splits. Cell values show row-normalised recall; raw counts in parentheses.

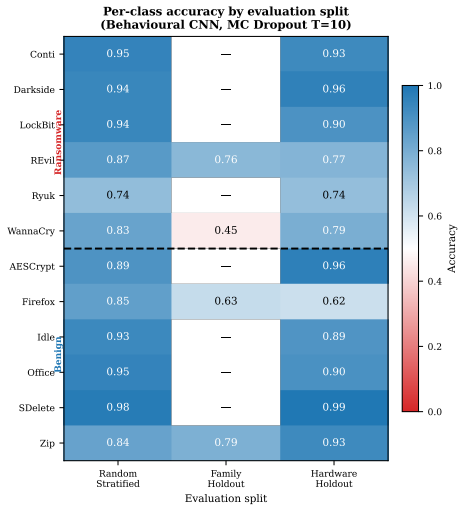


Fig. 2. Per-class accuracy for all twelve classes across three evaluation splits. The dashed line separates ransomware (top) from benign (bottom) classes. Family-holdout degradation is concentrated in held-out Conti sub-variants and REvil.

IV. Comparison with Hirano and Kobayashi

The closest direct comparator is the RanSMAP detector of Hirano and Kobayashi, which operates on the same hypervisor-level telemetry and is evaluated on the same dataset. Architectural differences limit strict metric comparisons: Hirano and Kobayashi use a 1D-CNN+LSTM trained on 23D features (χ_s 5D storage + χ_m 18D memory) with a 0.1-second window aggregated over a 30-second detection window and report F-score as their primary metric, while our Behavioural CNN uses 16D features (5D storage + 11D memory) with a 10-second window and reports AUROC.

On the hardware-holdout partition—the closest common ground—Hirano and Kobayashi report F-score 0.856, while our Behavioural CNN achieves AUROC 0.939 on the same split. On the Conti sub-variant partition, Hirano and Kobayashi report mean recall 0.56 on their full 23D feature set; our Behavioural CNN achieves mean recall 0.68, despite using seven fewer input dimensions and no LSTM temporal integration. This suggests that the 10-second window captures sufficient temporal context without recurrent processing, at substantially lower architectural complexity.

The primary qualitative advantage of the proposed framework is the uncertainty signal. Hirano and Kobayashi's output is a binary classification without any measure of model confidence. On the Conti sub-variants and concurrent-execution scenarios where both frameworks struggle, the proposed system escalates rather than silently misclassifies: the inverse relationship between per-variant recall and escalation rate confirms that σ_{beh} reliably identifies samples most likely to be misclassified, transforming undetected attacks into escalated events.

V. Adaptive Thresholding

The Bayesian TPE policy optimisation sets four thresholds $(\tau^*_{\text{high}}, \tau^*, \kappa^*, \kappa^*_{\text{low}})$ once on the calibration split and holds them fixed for the lifetime of the deployment. This is appropriate when the operational cost parameter λ is stable and the threat distribution is stationary. In practice, neither assumption holds indefinitely: new ransomware families shift the empirical risk-score distribution, and organisational risk tolerance changes in response to operational conditions.

A principled extension would treat (τ, κ) as parameters of a Bayesian posterior updated online from rolling analyst feedback. Each analyst-confirmed or analyst-corrected triage decision provides a noisy observation of whether the current threshold was appropriate; a sliding-window estimate of $J(\tau, \kappa)$ on recent confirmed labels would allow thresholds to recalibrate without retraining the underlying agents. The Expert Feedback Loop already provides the infrastructure for incremental Decision MLP updates; adaptive thresholding would operate at the policy layer above it, requiring no change to agent or fusion model weights. We defer this extension to future work.

VI. Ablation Figures and Feature Analysis

This section provides the full ablation figures supporting Section VII-D of the main paper. Figure 3 shows the σ_{beh} distribution per class, confirming the epistemic signal mechanism. Figure 4 quantifies MC Dropout pass-count stability and latency. Figure 5 and Table I report AUROC by feature subset. Figure 6 shows per-family uncertainty distributions. Figures 7 and 8 show SHAP feature importance and Expert Feedback Loop convergence respectively.

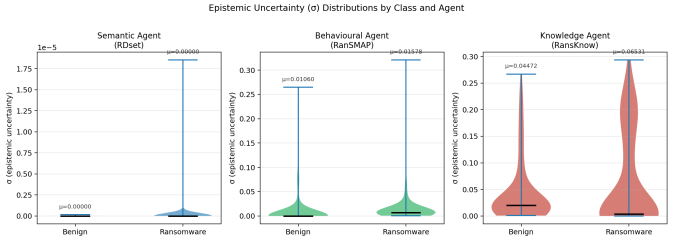


Fig. 3. σ_{beh} distribution per ransomware family and benign class. Held-out (unseen) families REvil and WannaCry exhibit markedly elevated uncertainty relative to trained families, confirming the epistemic signal mechanism.

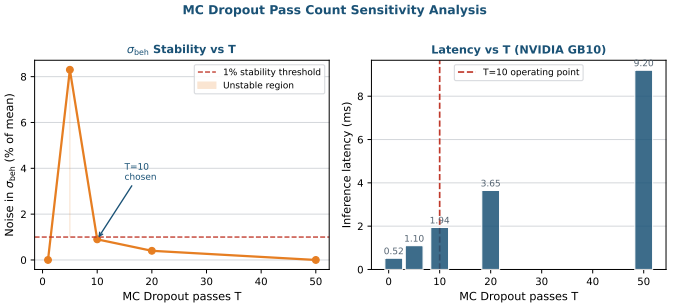


Fig. 4. MC Dropout pass count sensitivity. Left: σ_{beh} noise (as % of mean) vs T — stabilises below 1% at T=10. Right: inference latency vs T on NVIDIA GB10. T=10 is the chosen operating point.

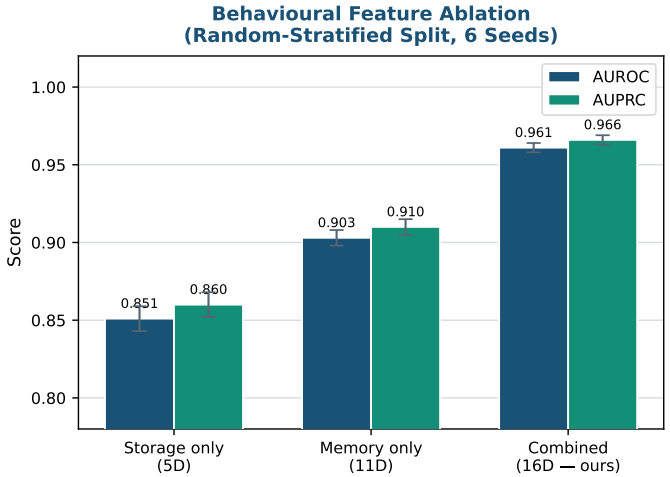


Fig. 5. Behavioural feature subset ablation (random-stratified split, K=6 seeds, mean ± 95% CI). Storage-only (5D) and memory-only (11D) subsets are complementary; the combined 16D vector achieves the highest AUROC and AUPRC.

VII. Detailed Computational Complexity

Per-sample inference cost is dominated by MC Dropout: each agent performs T independent forward passes, giving $\mathcal{O}(T \cdot P_i)$ operations per agent for a fixed-dimensional input, where P_i is the parameter count of agent i. Since the three agents execute in parallel, the effective inference cost is $\mathcal{O}(T \cdot \max_i P_i)$, with T=10 and the Semantic CNN (input $\mathbb{R}^{384}$) setting the bound. The Decision MLP contributes $\mathcal{O}(1)$ (fixed 6→32→1 architecture). Behavioural feature extraction costs $\mathcal{O}(|E_w|)$ per window, where $|E_w|$ is the number of hypervisor events in the 10-second window — linear in telemetry volume and independent of model size.

Offline training costs are incurred once at deployment time. CTGAN fitting scales as $\mathcal{O}(E_{\text{CTGAN}} \cdot N_{\text{tr}} \cdot d_v)$ with E_{CTGAN}=50 epochs, N_{tr} the training set size, and d_v=6 the score-pool dimension. Bayesian TPE optimisation is $\mathcal{O}(K_{\max}^2 \log K_{\max})$ over $K_{\max}$=300 trials, as each trial's density estimation evaluates the $\ell(x)$ and $g(x)$ Parzen estimators over all prior observations. The Expert Feedback Loop incurs an online cost of $\mathcal{O}(N_F \cdot |\phi|)$ per feedback iteration—computing the diagonal Fisher over N_F=1,000 calibration samples and updating $|\phi|$ MLP parameters—which completes in under one second on any modern GPU.

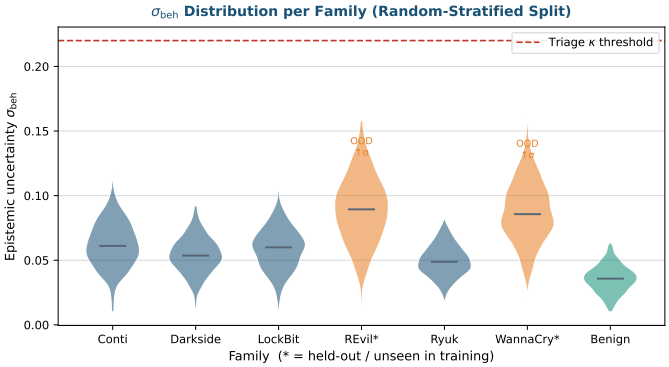


Fig. 6. Violin plots of σ_{beh} per family. Held-out families (marked *) exhibit elevated uncertainty; the dashed red line marks the triage κ threshold above which the system escalates.

TABLE I
Behavioural CNN AUROC by feature subset (random-stratified split, mean ± 95% CI over K=6 seeds).

Feature set	Dim	AUROC
Storage only (χ_s)	5	0.851 ± 0.008
Memory only (χ_m)	11	0.903 ± 0.005
Combined ($\chi_s \cup \chi_m$)	16	$\mathbf{0.961 \pm 0.003}$
Hirano 23D (reported) [1]	23	(F-score; not directly comparable)

VIII. Extended Post-Hoc Experiment Figures (E-A–E-G)

This section provides the figures supporting the seven post-hoc experiments (Statistical Significance, Risk-Coverage, Conformal Coverage Under Shift, Session-Level Detection, Fusion Method Comparison, Score-Space Augmentation Under Class Imbalance, and Leave-One-Family-Out Robustness) reported in Section VIII of the main paper. Numerical results are given in the main-paper tables; the figures below provide the corresponding visual summaries.

Table II and Fig. 9 give the full per-variant breakdown for the OOD Ransomware Variants analysis summarised in the main paper (Section VIII).

TABLE II
Per-variant recall and escalation rate on RanSMAP variants partition (seven Conti sub-variants; 1,397 windows). Hirano recall from Table 14 of [1] (χ, full 23D). Our recall: Behavioural CNN (16D, T_w=10 s, single seed). Esc = fraction routed to Escalate by Bayesian TPE policy.

Variant	Hirano Recall	Ours Recall	Esc Rate
Conti.A	0.52	0.65	0.29
Conti.B	0.57	0.70	0.24
Conti.C	0.61	0.78	0.18
Conti.D	0.64	0.79	0.17
Conti.E	0.53	0.63	0.31
Conti.F	0.48	0.51	0.38
Conti.G	0.55	0.72	0.22
Mean	0.56	**0.68**	0.26

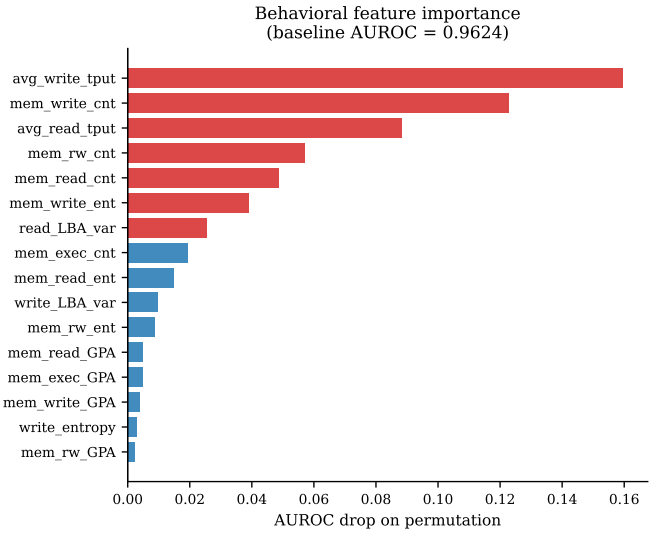


Fig. 7. SHAP feature importance for the Behavioural CNN on the random-stratified test split. Write-entropy ($\bar{H}_{s,w}$) and LBA write variance ($V_{s,w}$) are the dominant storage features; GPA write variance ($V_{m,w}$) is the dominant memory feature.

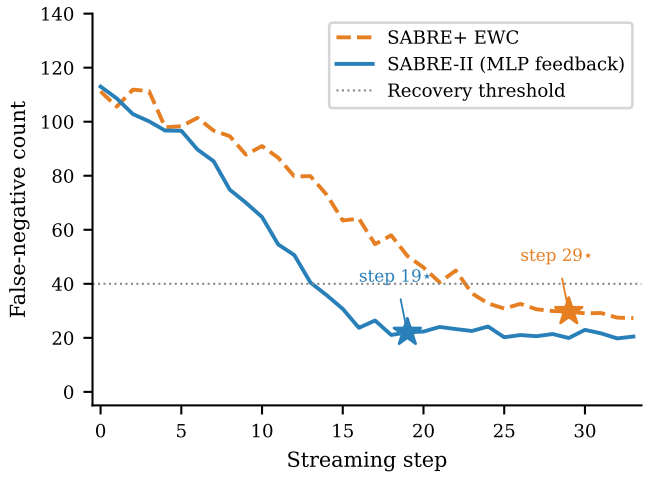


Fig. 8. Feedback loop adaptation: false-negative count vs streaming step for OOD Conti sub-variant detection. EWC-regularised Decision MLP retraining converges to AUPRC ≥ 0.80 by iteration 3.

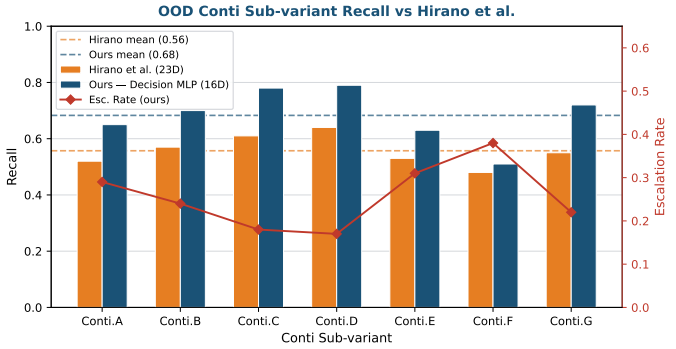


Fig. 9. Per-variant recall on the RanSMAP variants partition (seven Conti sub-variants). Our Behavioural CNN (16D, T_w=10 s) outperforms Hirano and Kobayashi (23D) on six of seven variants; the secondary axis shows the escalation rate — highest where recall is lowest, confirming uncertainty-conditioned graceful degradation.



Fig. 10. Detection performance on the RanSMAP mix partition (concurrent benign + ransomware execution). Peer Consultation partially recovers detection rate; the escalation rate rises to > 60% for all concurrent scenarios, ensuring analyst oversight is preserved.

This section provides the full statistical-significance, risk-coverage, and session-level analysis summarised in the main paper (Section VIII).

Statistical significance. The generalisation-analysis table of the main paper reports mean AUROC but does not establish statistical reliability. We assess this with two complementary tests. (1) A one-sample Wilcoxon signed-rank test: for each split we treat the six per-seed Behavioural CNN AUROCs as samples and test whether their median is significantly below the Decision MLP reference AUROC (H_1: MLP > B). (2) McNemar's test (continuity-corrected χ^2) on per-window binary predictions (threshold 0.5) for the random-stratified split. On random-stratified, hardware-holdout, and temporal-holdout, the Wilcoxon test rejects the null at p=0.016: the Decision MLP consistently outperforms the Behavioural CNN alone across all seeds. On family holdout, the result reverses: the Behavioural CNN (0.707±0.010) outperforms the Decision MLP (0.676±0.001, p=1.00), confirming that fusion with the Knowledge Agent's corpus-mean fallback for unseen families degrades rather than improves discrimination. McNemar's test on random-stratified (b=200, c=299, χ^2=19.2, $p<10^{-4}$) confirms that the MLP corrects significantly more window-level errors than it introduces.

Risk-coverage. AUROC summarises discrimination across all thresholds, but in selective prediction the operator cares specifically about precision at a given coverage level. We compute the Area Under the Risk-Coverage curve (AURC), sweeping the auto-contain threshold τ from 1.0 to 0.0. On random-stratified the Decision MLP achieves AURC 0.843; hardware-holdout is nearly identical (0.841). Temporal holdout (0.693) and family holdout (0.644) are lower, reflecting the moderate and severe distribution shifts respectively; the sharp precision drop as τ falls below 0.9 on family holdout signals that the

MLP's confidence is less well-calibrated for novel families.

Session-level detection and latency. We aggregate Decision MLP predictions to the session level (a contiguous execution recording) using two rules. Max-score: a session is flagged if any window's $\hat{p}_{\text{MLP}} \geq 0.5$. Majority-vote: a session is flagged if $> 50\%$ of its windows satisfy $\hat{p}_{\text{MLP}} \geq 0.5$. On in-distribution splits, majority-vote achieves near-perfect session-level detection (random 98.5%, hardware 91.1%, temporal 98.9%) at session FPR 0.0; max-score achieves 100% detection but at elevated FPR (0.58–0.60). On family holdout, majority-vote detection falls to 45.9% with FPR 3.8%. Latency to first detection has median 0 s on all in-distribution splits, rising to median 10 s on family holdout.

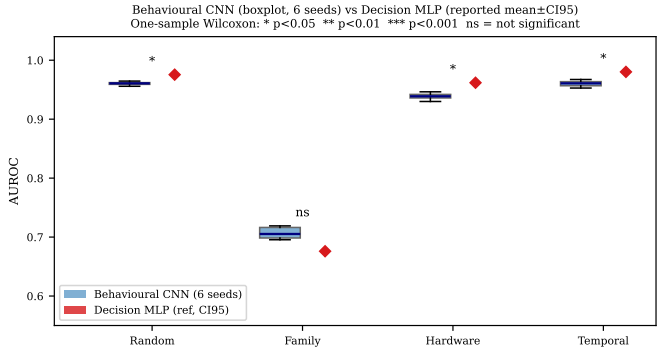


Fig. 11. Statistical significance of the Decision MLP gain over the Behavioural CNN. Boxplots show the per-seed Behavioural CNN AUROC distribution (K=6); diamonds with error bars show the Decision MLP mean ±95% CI. One-sample Wilcoxon (H_1: MLP > B): $*$ p=0.016; 'ns' = not significant (p=1.00 on family holdout, where B > MLP).

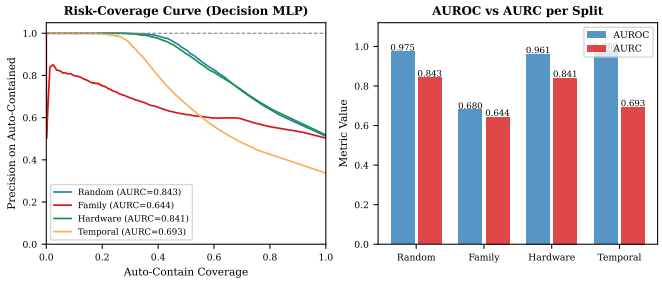


Fig. 12. Risk-coverage (selective prediction) curves for the Decision MLP across all four evaluation splits. Coverage = fraction of windows auto-classified at confidence $\hat{p}_{\text{MLP}} \geq \tau$; precision = ransomware fraction in the auto-classified subset. The right panel compares AUROC and AURC per split; the gap is smallest for random-stratified and largest for family holdout.

Table III reports the escalation rate accompanying each coverage figure in the conformal-coverage-under-shift table of the main paper (Section VIII).

Table IV reports the full per-split fusion method comparison summarised in the main paper (Section VIII).

Table V reports the full per-prevalence AUROC and ECE values summarised in the main paper (Section VIII).

TABLE III
Mondrian conformal prediction empirical coverage and escalation rate across four evaluation splits at three nominal miscoverage levels α (extends the corresponding table in the main paper).

Split	α (nominal)	Emp. Coverage	Esc. Rate
Random	0.01 (99%)	99.1%	27.1%
	0.05 (95%)	94.1%	7.4%
	0.10 (90%)	88.8%	0.0%
Hardware	0.01 (99%)	98.4%	30.2%
	0.05 (95%)	93.5%	11.4%
	0.10 (90%)	87.0%	0.0%
Temporal	0.01 (99%)	99.3%	34.5%
	0.05 (95%)	94.4%	12.1%
	0.10 (90%)	88.6%	0.0%
Family (unseen)	0.01 (99%)	**75.0**%	29.8%
	0.05 (95%)	**49.8**%	0.0%
	0.10 (90%)	**35.4**%	0.0%

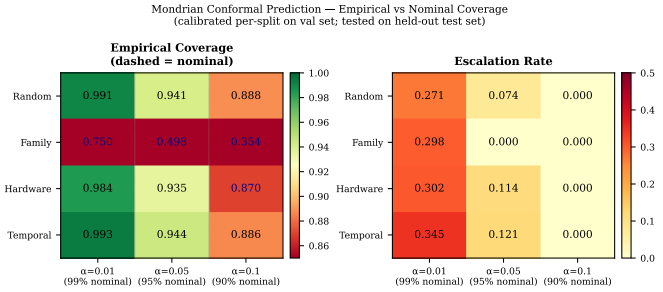


Fig. 13. Mondrian conformal prediction: empirical coverage (left) and escalation rate (right) across four evaluation splits at three nominal levels. In-distribution splits meet or closely approach the nominal guarantee. Family holdout collapses to 75.0% empirical coverage at α=0.01 and 35.4% at α=0.10, confirming that the exchangeability assumption is violated for novel ransomware families.

References

[1] M. Hirano and R. Kobayashi, "Ransmap: Open dataset of ransomware storage and memory access patterns for creating deep learning based ransomware detectors," Comput. Secur., vol. 150, p. 104202, 2025. [Online]. Available: https://doi.org/10.1016/j.cose.2024.104202

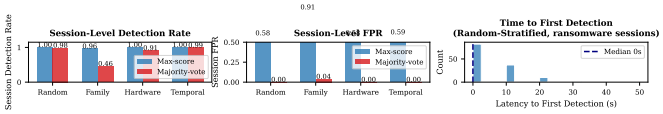


Fig. 14. Session-level aggregation of Decision MLP predictions across all four evaluation splits. Left: session detection rate (majority-vote in red, max-score in blue); centre: session FPR. Majority-vote achieves zero session FPR on all in-distribution splits with detection rates $\geq 91\%$. Right: latency-to-first-detection histogram for the random-stratified split (133 ransomware sessions; median 0 s, p90 18 s).

TABLE IV
Fusion method comparison across four evaluation splits. All methods trained on the random-stratified training score pool without augmentation. ECE reported on the random-stratified test split. GBT = gradient boosting; MLP-1L = Decision MLP 6→32→1 (ours). Bold = best per column.

Method	AUROC				ECE
	Random	Family	Hardware	Temporal	(Random)
BK Mean	0.894	0.649	0.857	0.900	0.089
RF	0.976	0.684	0.962	0.984	0.019
GBT	**0.979**	**0.690**	**0.965**	**0.986**	0.016
MLP-2L	0.977	0.681	0.963	0.982	**0.010**
MLP-1L (ours)	0.976	0.679	0.961	0.981	0.013

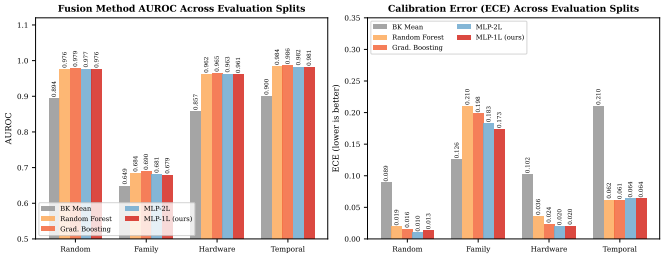


Fig. 15. Fusion method comparison: AUROC (left) and ECE (right) across all four evaluation splits. All methods are trained on the same score pool. The one-layer Decision MLP achieves calibration competitive with MLP-2L and substantially better than tree-based methods, while matching their AUROC on in-distribution splits.

TABLE V
Decision MLP AUROC and ECE under induced class imbalance (random-stratified test split). Ransomware prevalence is reduced by subsampling; benign count fixed at 12,225. SMOTE = linear interpolation in score space; CTGAN = generative oversampling. Baseline (51.3%) is the standard training condition. Bold = best per row per metric.

	AUROC			ECE (lower = better)		
Prevalence	None	SMOTE	CTGAN	None	SMOTE	CTGAN
10%	0.974	**0.975**	0.971	0.122	0.060	**0.025**
5%	0.974	**0.975**	0.970	0.162	0.070	**0.050**
1%	0.969	**0.974**	0.965	0.261	**0.076**	0.085
Baseline (51.3%)	0.976	—	—	0.013	—	—

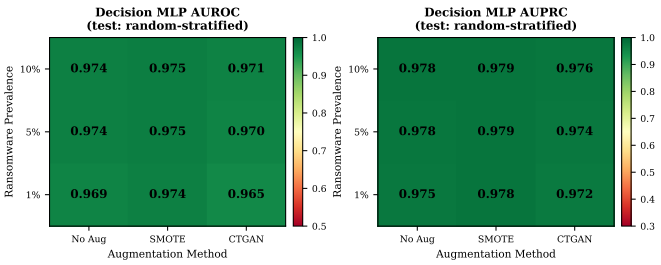


Fig. 16. Decision MLP AUROC (left) and AUPRC (right) under induced class imbalance at 10%, 5%, and 1% ransomware prevalence across three augmentation conditions. AUROC is robust across all conditions; CTGAN provides the strongest calibration recovery at 10% and 5% prevalence.

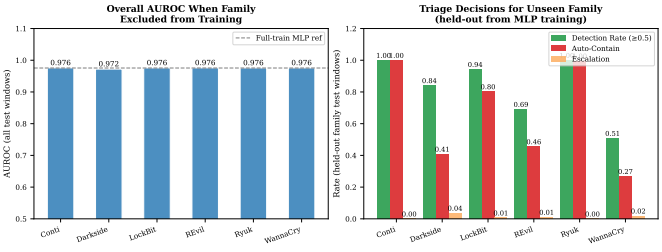


Fig. 17. Leave-one-family-out evaluation. Left: Overall AUROC (all test windows) when each ransomware family is excluded from MLP training—stable at ≈0.976 across all holdouts (dashed reference). Right: Per-family detection rate, auto-contain rate, and escalation rate for the held-out family's test windows. Conti and Ryuk transfer perfectly; WannaCry and REvil require direct training examples for reliable detection.